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AI-POWERED RESTAURANT MANAGEMENT SYSTEM WITH AN INTELLIGENT CHATBOT (MERAKE PIZZERIA CAFE)

BCC3024 UNDERGRADUATE PROJECT II



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INTRODUCTION

MerakiDineIn is an AI-powered restaurant management system that combines intelligent automation, comprehensive analytics, and seamless customer service. The platform features an AI chatbot for customer assistance, smart booking management, real-time business intelligence, and integrated payment processing, all of which are designed to optimize restaurant operations and enhance the overall dining experience.

PROBLEM STATEMENT

- The absence of a centralized digital system limits the café’s ability to meet the expectations of digital-native customers who prefer online booking and pre-ordering.
- The café currently relies on manual reservation methods (walk-ins and phone calls), which are time-consuming, error-prone, and inefficient.
- Small and medium-sized cafes such as Meraki Pizzeria are unable to afford custom booking systems due to high development and maintenance costs.

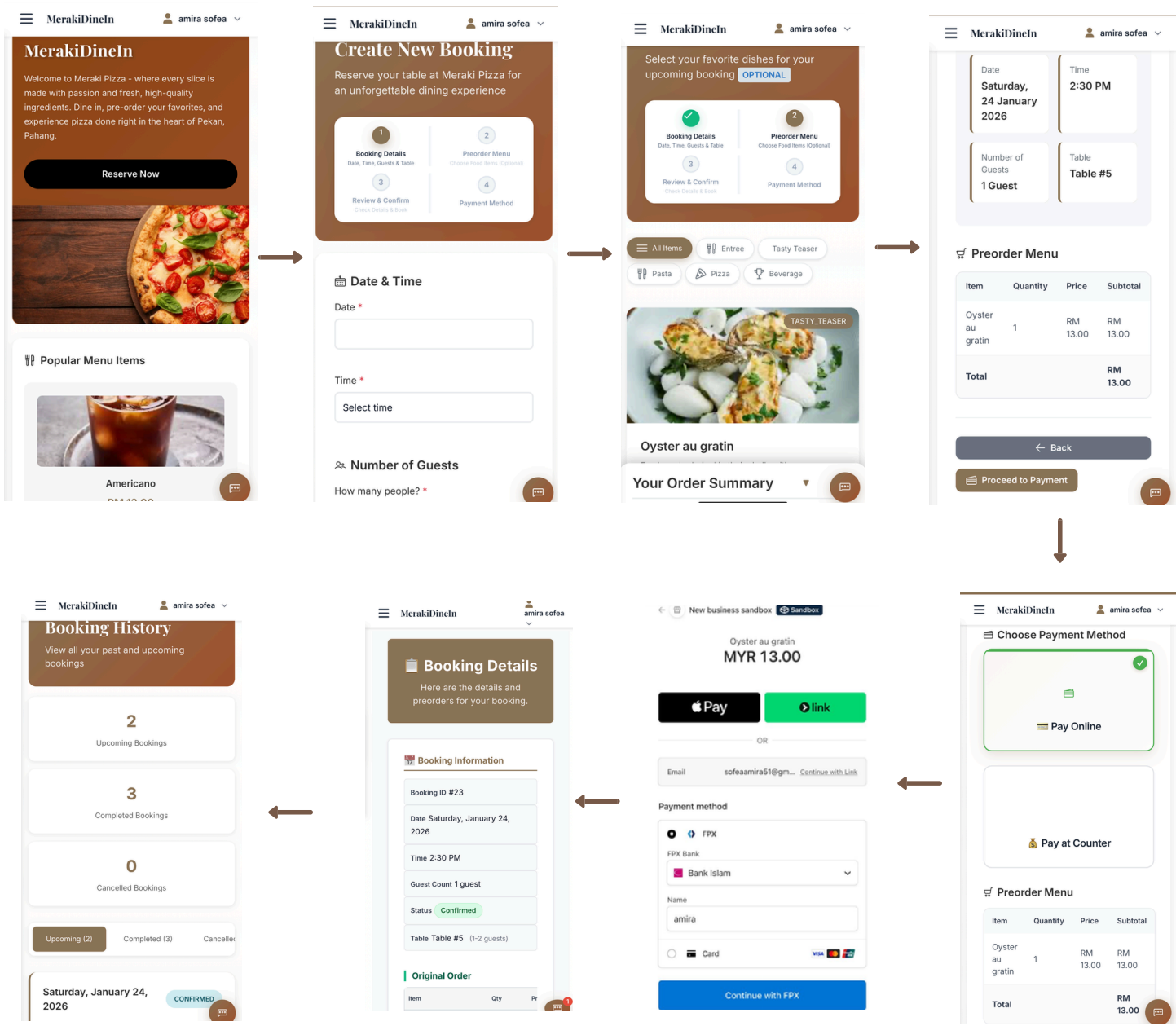
OBJECTIVES

- To analyze system requirements through surveys conducted with Meraki Pizzeria customers and the cafe owner.
- To develop a web-based cafe reservation system with an AI chatbot for bookings, pre-orders, and inquiries.
- To evaluate the system’s effectiveness in improving cafe operations and customer experience through user feedback.

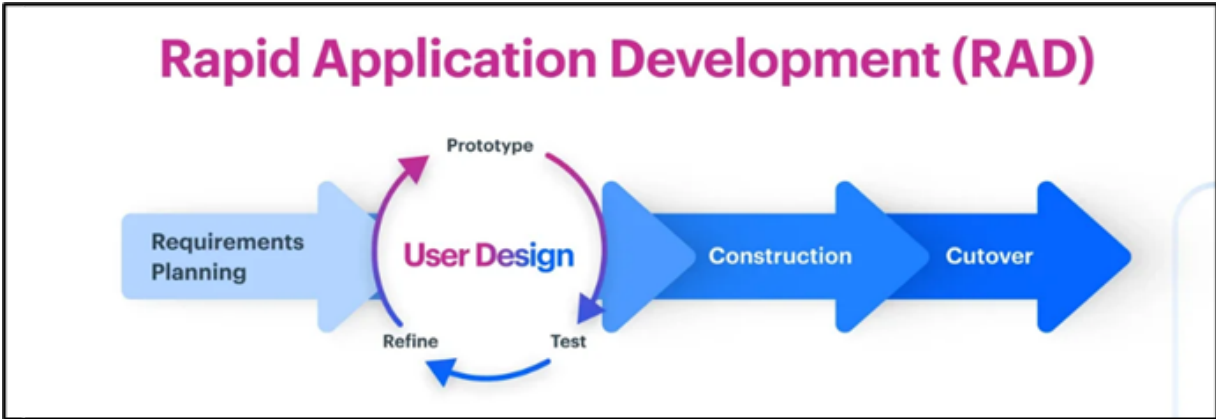
SCOPES

- Designed exclusively for Meraki Pizzeria (single cafe system).
- Supports dine-in reservations only (no takeaway or delivery).
- Includes three user roles: customers, staff, and administrators, with role-based access.

RESULTS



METHODOLOGY



MARKETABILITY

- The MerakiDineIn system can be integrated with POS systems, online payment gateways, inventory systems, and third-party services, making it suitable for small and medium-sized cafes or F&B organizations.

CONCLUSION

The Meraki Pizzeria Booking System successfully provides a cost-effective and efficient solution for managing café reservations and food pre-orders. The system enhances operational efficiency, reduces booking errors, and improves the overall customer experience, making it suitable for adoption by small and medium-sized cafes.